JAVA PROGRAM REPORT

Hybrid Inheritance Using Interfaces

Programming Language:	Java
Topic:	Hybrid Inheritance
Program File:	HybridInheritance.java

Objective

To demonstrate hybrid inheritance in Java by combining multilevel and hierarchical inheritance. SportsStudent inherits from Student, which inherits from Person, while Employee also inherits from Person.

Java Source Code

```
class Person {
    void showPerson() {
        System.out.println("This is a person");
    }
}

class Student extends Person {
    void study() {
        System.out.println("Student studies");
    }
}

class SportsStudent extends Student {
    void play() {
        System.out.println("Student plays sports");
    }
}

class Employee extends Person {
    void work() {
        System.out.println("Employee works");
    }
}

public class HybridInheritance {
    public static void main(String[] args) {

        SportsStudent s = new SportsStudent();
        s.showPerson();
        s.study();
        s.play();

        Employee e = new Employee();
        e.showPerson();
        e.work();
    }
}
```

Program Explanation

1. Person Class: The base class defines showPerson(), which prints a common message.

2. Student Class: Student extends Person and adds the study() method.

3. SportsStudent Class: SportsStudent extends Student, so it inherits showPerson() and study(). It adds play().

4. Employee Class: Employee independently extends Person and adds the work() method.

5. Hybrid Inheritance: The program combines hierarchical inheritance (Student and Employee both extend Person) with multilevel inheritance (Person → Student → SportsStudent).

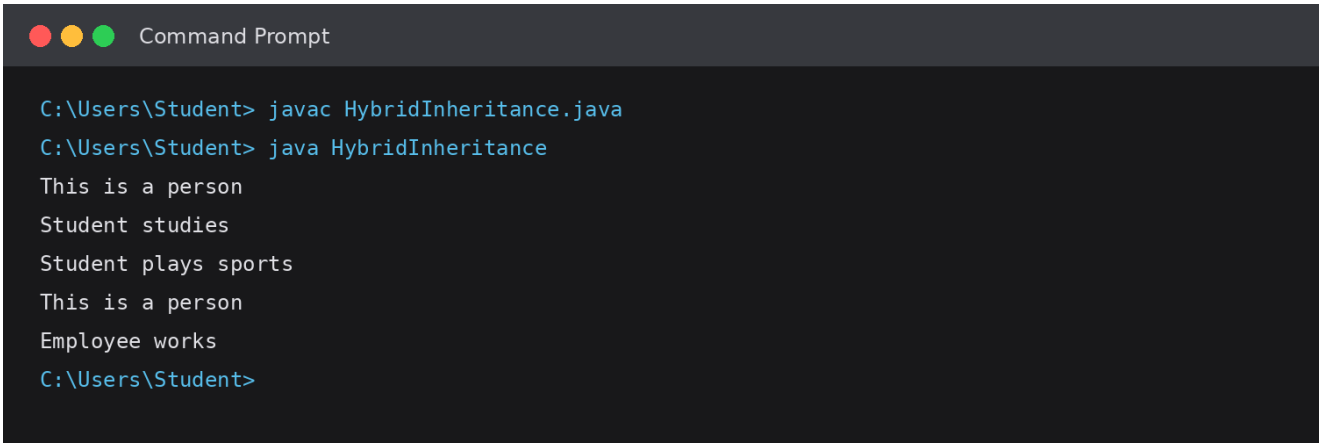
6. Main Method: The program creates a SportsStudent object and an Employee object, then calls inherited and class-specific methods.

Expected Output

```
This is a person
Student studies
Student plays sports
This is a person
Employee works
```

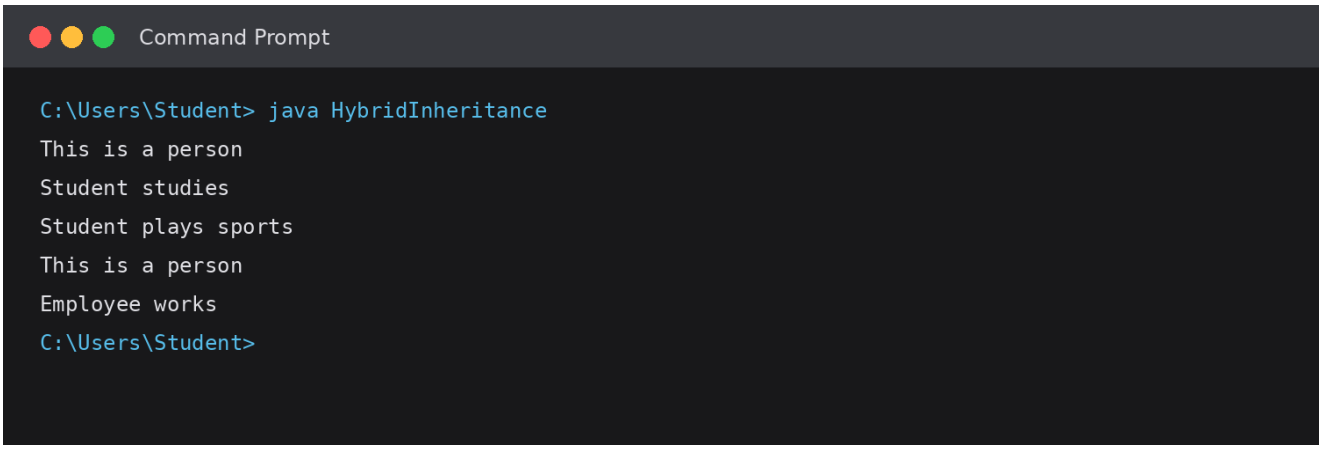
Execution and Output Screenshots

Test Case 1: Compilation and Execution



Verification: The expected messages are displayed for the SportsStudent and Employee objects.

Test Case 2: Re-execution Verification



Verification: Re-running the program produces the same output sequence.

Test Case Summary

Test Case	Expected Result	Actual Result	Status
Test Case 1	SportsStudent and Employee methods execute.	Output matches expected messages.	**PASS**
Test Case 2	Same output on re-execution.	Output remains consistent.	**PASS**

Conclusion

The program demonstrates hybrid inheritance by combining hierarchical and multilevel inheritance. SportsStudent inherits behavior through Student from Person, while Employee also inherits from Person. The output shows that inherited and specialized methods execute as expected.